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PAPER

Quantum utility-scale error mitigation for quantum quench dynamics in Heisenberg spin chains

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E-mail: kyu@bnl.gov**Keywords:** quantum error mitigation, heisenberg spin chain, trotterization, quantum utility, entanglement entropy**Abstract**

We implement a quantum error mitigation method termed self-mitigation, which is comparable to zero-noise extrapolation, at large scales to achieve quantum utility on near-term, noisy quantum computers. We investigate the effectiveness of several quantum error mitigation strategies, including self-mitigation, by simulating quantum quench dynamics for Heisenberg spin chains with system sizes up to 104 qubits using IBM quantum processors. In particular, we discuss the limitations of zero-noise extrapolation and the advantages offered by self-mitigation at large scales. The self-mitigation method demonstrates stable accuracy with large systems of 104 qubits comprising more than 3,000 CNOT gates. Also, we combine the discussed quantum error mitigation methods with practical entanglement entropy measuring methods, and it shows a good agreement with the theoretical estimation. Our study illustrates the usefulness of near-term noisy quantum hardware in examining the quantum quench dynamics of many-body systems at large scales and lays the groundwork for surpassing classical simulations with quantum methods prior to the development of fault-tolerant quantum computers.

1. Introduction

In recent years, unprecedented advancements in quantum computing have occurred, involving both quantum hardware and quantum algorithms. In fact, preliminary experiments on emerging quantum hardware indicate that these systems can outperform their classical counterparts for certain specialized computational tasks. However, demonstrating a comprehensive, significant, and precisely measurable quantum advantage over classical systems remains a formidable challenge, especially when tackling problems of practical relevance. One of the most challenging obstacles is the quantum noise originating from quantum devices. To overcome this, many quantum error correction (QEC) methods have been developed, and several roadmaps to fault-tolerant quantum computing (FTQC), for which QEC is crucial, have been suggested [1–3]. Even with an optimistic outlook, an early FTQC may be available by 2029. A recent study showed that 1457 physical qubits (distance 27 surface code) are required to achieve one logical qubit with 10^{-6} logical error rate (LER) [2]. This shows that reaching FTQC, even the early FTQC era, is quite demanding, despite rapid advances in quantum computing and considering current in-service quantum computers have less than 200 qubits. Therefore, it is essential to maximize the efficiency of today's noisy quantum computers in the pre-FTQC phase and in the early FTQC systems.

On the other hand, simulating fundamental physics with near-term, noisy quantum computers presents major challenges, including high error rates that impact computational accuracy, limited qubit counts that restrict the complexity of systems that can be simulated, and difficulties in maintaining qubit coherence over time. Nonetheless, advances in error mitigation (QEM) methods [4–11] are steadily improving the ability of

noisy quantum devices to carry out detailed and accurate simulations of fundamental physics. These developments have demonstrated the practical utility of noisy quantum computers even before the arrival of fully fault-tolerant quantum systems [11]. However, simulating large quantum systems and accurately extracting observables for more realistic problems on near-term quantum computers still remains out of reach for now. In addition to measuring observables, measuring the entanglement entropy is also crucial because it provides deep insight into the quantum correlations present in a system and serves as a powerful diagnostic tool in many areas of quantum physics. In non-equilibrium quantum systems, entanglement entropy growth (e.g., after a quench) reflects how quantum information spreads. Therefore, measuring the entanglement entropy is key to understanding thermalization, many-body localization (MBL), and scrambling of quantum information. However, measuring the entanglement entropy on near-term quantum computers is still challenging. Addressing the challenges of measuring observables and entanglement entropy requires further exploration to better understand the practical capabilities and limitations of current quantum hardware for large-scale quantum simulations.

Therefore, it is essential to efficiently apply QEM methods to the quantum algorithms, as well as develop more efficient QEM techniques, to enable the effective use of near-term noisy quantum computers at the utility scale, before the advent of fully FTQC, even in its early stages. Hence, we advance a quantum error mitigation method termed self-mitigation [12], which is comparable with zero-noise extrapolation [5, 6, 13] to achieve quantum utility on near-term, noisy quantum computers. We implement the self-mitigation method to the optimized second-order Trotterization [14] for the XXZ spin chain model. We evaluate several quantum error mitigation techniques, including self-mitigation, and analyze their respective strengths and weaknesses. In particular, we analyze the limitations of zero-noise extrapolation and the advantages offered by self-mitigation at a large scale. Our results demonstrate that self-mitigation surpasses zero-noise extrapolation in both accuracy and resource efficiency. Building on this benchmark analysis (20-qubit system), we extend our experiments to larger systems, scaling up to 84 qubits with periodic boundary conditions and 104 qubits with open boundary conditions. The extended simulations show a good accuracy with more than 3,000 CNOT gates. The self-mitigation method shows stable accuracy with the large systems, while the zero-noise extrapolation method has a higher error rate than the benchmark test of 20-qubits. Finally, we explore computing the entanglement entropy after time evolution of the Hamiltonian. We address a practical protocol for the entropy measurement on quantum devices and implement it on IBM quantum computers. The entropy measurement result shows a good agreement with the classical result.

We focus on studying the quench dynamics in the one-dimensional anisotropic spin- $\frac{1}{2}$ antiferromagnetic Heisenberg model or XXZ spin chain using IBM superconducting quantum computers and assess their ability to accurately capture the intricate spin dynamics of the model. The XXZ model is a paradigmatic example of a quantum spin system that exhibits integrability and critical behavior [15–18] where the critical regime of the XXZ spin chain can be described by a $1 + 1$ D conformal field theory with central charge $c = 1$ [19, 20]. Besides, the low-energy limit of the XXZ spin chain can be connected to lower-dimensional quantum field theories, as shown in [21, 22]. In addition, the time evolution in the XXZ spin chain after the quantum quenches at zero temperature is studied in [23–25] which focused on dynamics of Von Neumann entropy of a block of spins and spin-spin correlation functions after a sudden quench in the anisotropy parameter, and for finite temperature case in [26] which studied the spreading of information after quenches from a variety of initial thermal density matrices. For a detailed discussion on quench dynamics and relaxation in isolated integrable quantum spin chains, refer to [27]. In a recent study, [28] reported the preparation of the antiferromagnetic ground state of a 50-site Heisenberg XXZ spin chain near the XY phase boundary, followed by the study of its dynamics on an IBM quantum computer.

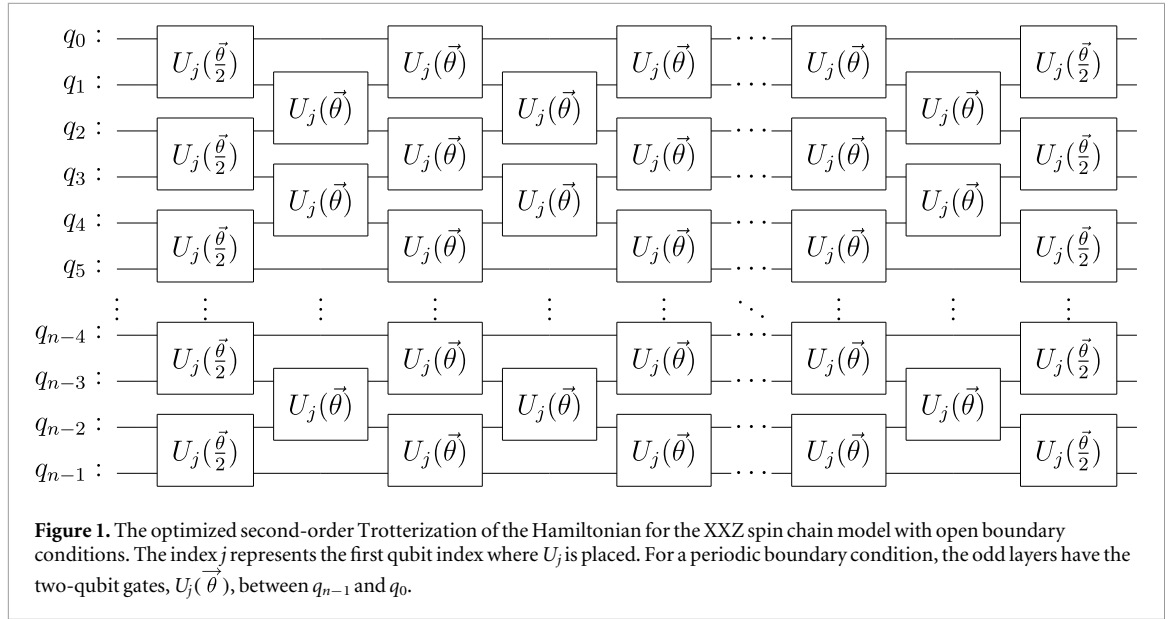
The following section reviews the quantum circuit implementation of the XXZ spin chain Hamiltonian used in this study. Section 3 presents an overview of the quantum error mitigation techniques used in this study, particularly emphasizing our self-mitigation implementation in section 3.5. In section 4, we compare these methods and evaluate their effectiveness. Finally, we evaluate the entanglement entropy of the quantum state after its time evolution governed by the Hamiltonian.

2. Implementation of time evolution for the XXZ spin chain hamiltonian

The Heisenberg XXZ spin chain model is described as follows:

$$H = J_1 \sum_{i=1}^N (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z) \quad (1)$$

where the antiferromagnetic nearest-neighbor coupling $J_1 > 0$ and the exchange-anisotropy parameter Δ controls the parameter space of the Hamiltonian. The system exhibits a ferromagnetic phase when $\Delta < -1$, a



critical phase when $-1 < \Delta < 1$, and an antiferromagnetic phase when $\Delta > 1$. We adopt a value of $\Delta = 0.5$ for our experiments, as this situates the model in a critical phase. Besides, the spin operators, $S^i = \frac{1}{2}\sigma^i$ obey the $SU(2)$ algebra,

$$[S_i^\alpha, S_j^\beta] = i\delta_{ij}\epsilon^{\alpha\beta\gamma}S_i^\gamma,$$

where $\alpha, \beta, \gamma = x, y, z$ and $i, j = 1, \dots, N$. Equation (1) is reformulated by the Pauli operators, $\sigma_j^x, \sigma_j^y, \sigma_j^z$ as follows:

$$H = \frac{J_1}{4} \sum_{j=0}^{N-1} (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta \sigma_j^z \sigma_{j+1}^z). \quad (2)$$

To execute the time evolution driven by the Hamiltonian in equation (2), we use the second-order Trotterization [29–31] as follows:

$$U_{tot}(t) = e^{-iHt} \approx \left(\prod_{n=1}^M e^{-iH_n t/2r} \prod_{n=M}^1 e^{-iH_n t/2r} \right)^r, \quad (3)$$

where $H = \sum H_n$ and r is the number of Trotter steps. The Trotter timestep size is defined as $\delta t = t/r$. We adopt an efficient second-order Trotterization for the model that is described in [14]. The circuit diagram is visualized in figure 1.

The second-order Trotterization is composed of the following basic building blocks.

$$U_j(\vec{\theta}) = \exp \left(-i \left(\frac{\theta_x}{2} \sigma_j^x \sigma_{j+1}^x + \frac{\theta_y}{2} \sigma_j^y \sigma_{j+1}^y + \frac{\theta_z}{2} \sigma_j^z \sigma_{j+1}^z \right) \right), \quad (4)$$

where $\vec{\theta} = (\theta_x, \theta_y, \theta_z) = (\frac{1}{2}\delta t, \frac{1}{2}\delta t, \frac{1}{2}\Delta\delta t)$ with the Trotter step size δt acting on qubit j and $j+1$. By the Trotter approximation, we approximate the time evolution $U_{tot}(t)$ in equation (3) by arranging the $U_j(\vec{\theta})$ operators in staggered placement (brick-wall structure) [32, 33] and finally optimized as shown in figure 1. Note that the optimized second-order Trotterization has only $2M+1$ layers of U_j building blocks with M Trotter steps, while the first-order Trotterization has $2M$ layers with M Trotter steps. Hence, the optimized second-order Trotterization is implemented by adding one even layer at the end of the first-order Trotterization and adjusting the angle parameters $(\vec{\theta})$.

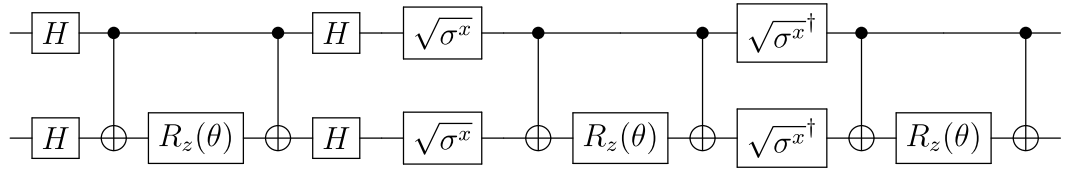
2.1. Unit block implementation

Since $U_j(\vec{\theta})$ in equation (4) is the basic building block for the quantum circuit implementation as discussed in the previous section, achieving reliable simulations on noisy quantum computers critically depends on the design and implementation of efficient quantum circuits, given that near-term noisy devices are highly susceptible to multiple sources of quantum noise, particularly gate errors.

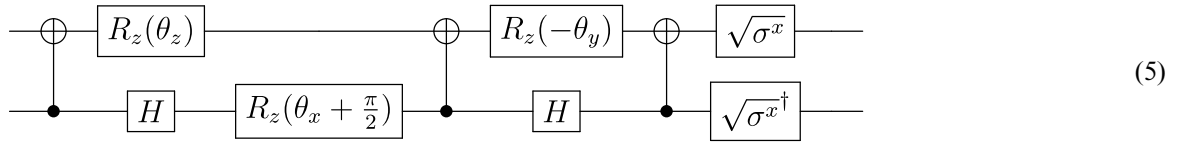
Based on RZZGate in IBM Qiskit, we construct the induced $R_{X_i X_j}(\theta)$ and $R_{Y_i Y_j}(\theta)$ gates by the gate identities as follows:

$$\begin{aligned}
R_{XX}(\theta) &= \begin{array}{c} \text{---} H \text{---} \bullet \text{---} \bullet \text{---} H \text{---} \\ | \quad | \\ \text{---} H \text{---} \oplus \text{---} R_z(\theta) \text{---} \oplus \text{---} H \text{---} \end{array}, \\
R_{YY}(\theta) &= \begin{array}{c} \text{---} \sqrt{\sigma^x} \text{---} \bullet \text{---} \bullet \text{---} \sqrt{\sigma^{x\dagger}} \text{---} \\ | \quad | \\ \text{---} \sqrt{\sigma^x} \text{---} \oplus \text{---} R_z(\theta) \text{---} \oplus \text{---} \sqrt{\sigma^{x\dagger}} \text{---} \end{array}, \\
R_{ZZ}(\theta) &= \begin{array}{c} \bullet \quad \bullet \\ | \quad | \\ \oplus \text{---} R_z(\theta) \text{---} \oplus \end{array}
\end{aligned}$$

where H is the Hadamard gate, $\sqrt{\sigma^x} = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$, and $R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$. Hence, $U_i(\vec{\theta})$ in equation (4) is implemented as follows:



and this implementation has six CX gates and thirteen circuit depths. This circuit is compressed and optimized by circuit identities as follows:



and this circuit has three CX gates and seven depths. The derivation of the circuit identity is detailed in appendix A of [34]. Given that quantum gates are prone to errors, particularly two-qubit gates like CX, which tend to be significantly noisier than single-qubit gates, minimizing both the number of CX gates and the overall circuit depth is crucial for mitigating noise in quantum computations. Therefore, we utilize the quantum circuit defined in equation (5) for the implementation of equation (4).

3. Quantum error mitigations

A major challenge in running quantum algorithms on contemporary quantum devices, such as IBM Quantum processors, is the presence of errors and noise. To address these issues, researchers have proposed quantum error correction (QEC) techniques. However, QEC requires a significant qubit overhead, making it impractical to implement for large-scale problems on near-term noisy quantum processors, even with optimizations [35, 36]. In contrast, quantum error mitigation (QEM) embraces the imperfections of current quantum devices and employs techniques to reduce or suppress quantum errors and noise. Unlike QEC, QEM typically incurs little to no qubit overhead. In recent years, a variety of QEM methods have been developed and demonstrated to be effective in solving practical problems [9–11, 14, 37]. To mitigate quantum device errors and noise in our experiments, we employ five QEM techniques: Twirled Readout Error Extinction (TREX), Dynamical Decoupling (DD), Pauli Twirling (PT), Zero-Noise Extrapolation (ZNE), and Self-Mitigation (SM). The details of each QEM method are provided in the following sub-sections.

3.1. Twirled readout error extinction

Twirled Readout Error Extinction (TREX) is a technique used to mitigate readout errors in quantum circuits without measuring error rate explicitly [38]. It randomly applies Pauli-X gates to qubits just before measurement and then flips the corresponding classical bits in the output. In an ideal, noise-free setting, this process has no effect—it is effectively the identity. However, under realistic noise conditions, it symmetrizes the readout error channel, transforming complex, state-dependent bias into a simpler, uniform scaling factor.

TREX transforms arbitrary readout errors into a diagonal form by randomizing the measurement basis. This makes the error model much simpler (each qubit's error is independent and only affects its own measurement outcome), allowing for straightforward inversion and correction. Also, TREX does not require detailed modeling of the underlying noise, making it robust even when the noise is hard to characterize. However, it is less effective when the measurement errors are strongly correlated across qubits. In our experiments, we adopt 10 sampling numbers for the TREX.

3.2. Dynamical decoupling

Dynamical Decoupling (DD) is a method used to mitigate errors originating from unwanted interactions with neighboring or spectator qubits. It achieves this by periodically applying a sequence of control pulses that refocus the system's phase evolution, effectively averaging out the coupling between the qubits and their environment [39]. In practice, DD introduces single-qubit gate operations into idle intervals through suitable basis rotations, thereby shielding the qubits from correlated environmental noise. This process enhances the circuit's overall coherence time and stability. The performance of DD has been experimentally demonstrated across diverse quantum hardware platforms [10, 11, 37, 40, 41]. In this work, the DD sequence is implemented as $(t/4, X, t/2, X, t/4)$ during each idle period, where X indicates the `XGate` operation and t refers to the total idling time excluding the two `XGate` pulse durations.

3.3. Pauli twirling

Pauli Twirling (PT) is a noise randomization approach that mitigates coherent errors by converting them into stochastic Pauli-type errors. This is achieved by averaging out non-diagonal elements in the Pauli operator basis, $\{I, \sigma^x, \sigma^y, \sigma^z\}$, thereby simplifying the noise model to a depolarizing form [42–44]. Operationally, PT surrounds each Clifford gate with randomly chosen Pauli operators applied both before and after the gate, ensuring that the overall transformation remains mathematically equivalent to the ideal operation. Previous studies have experimentally verified the practicality and effectiveness of PT across multiple quantum hardware platforms [10, 11, 14, 45–47]. In this work, we restrict the application of PT to two-qubit Clifford gates, namely `CX` and `CZ`, where coherent noise is typically more prominent. To implement this, ten randomized versions of the base circuit are constructed, each incorporating distinct Pauli-twirling combinations selected from a predefined gate set, and the final expectation values are obtained by averaging over all circuit instances.

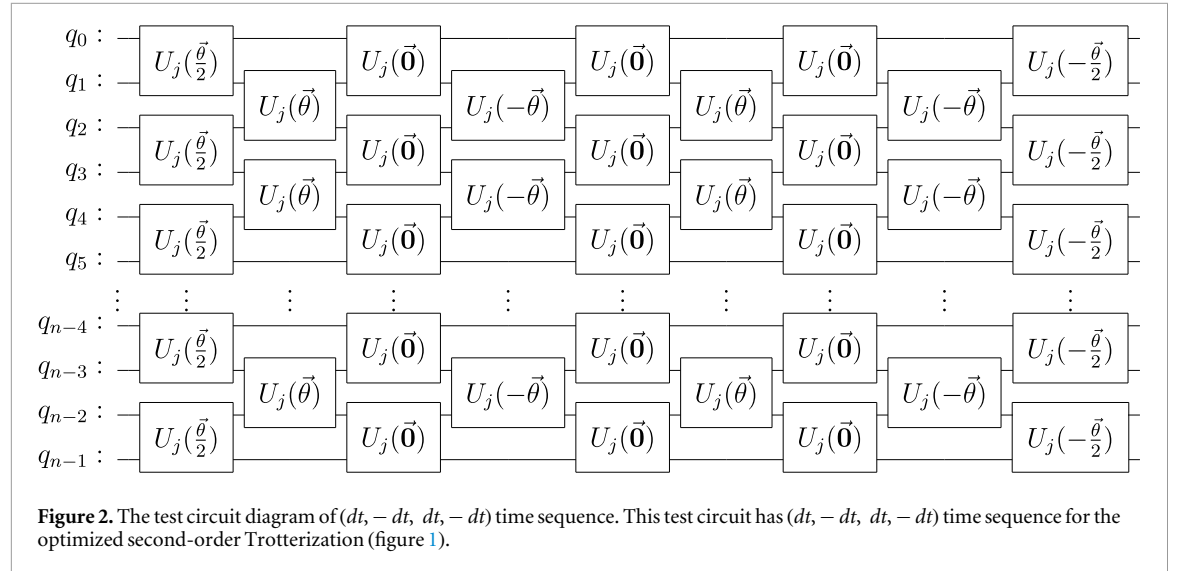
3.4. Zero-noise extrapolation

Zero-Noise Extrapolation (ZNE) mitigates quantum errors by inferring the noise-free expectation value through extrapolation from a set of noisy measurements collected at different noise strengths [5, 6, 13]. For our experiments, local unitary gate folding [13] is selectively applied to the two-qubit gates, `CX`, using scaling factors of 1, 3, and 5, given that the two-qubit gates are more than an order of magnitude noisier than single-qubit gates. Although various scaling factors are theoretically possible, the scaling factors of 1, 3, and 5 are essentially the only practical options on IBM quantum computers. When adopting a 7-fold scaling, the `CX` circuit depth (i.e., the number of `CX` layers) increases substantially and quickly exceeds the coherence limit. Consequently, a 7-fold scaling severely restricts the achievable circuit depth for implementing quantum algorithms. Conversely, using only scaling factors of 1 and 3 is insufficient to obtain a reliable quadratic or exponential fitting curve, since such extrapolation functions require at least three scaling points. Therefore, 1, 3, and 5 represent the practical maximum scaling factors available on current IBM quantum devices.

ZNE relies on the assumption that noise levels can be systematically increased in a controlled manner, despite the potential nonlinearity of the extrapolation curve. However, in reality, scaling the noise doesn't always produce predictable changes in the output. Furthermore, ZNE remains a heuristic approach, as the selection of its extrapolation parameters lacks a rigorous theoretical foundation. If the noise model is inaccurate or unstable, the extrapolation can become unreliable or even misleading. When the exact value is known, we can construct an accurate fitting curve for extrapolation. This curve, derived from small-scale estimations, can then be extended to larger scales under the assumption that the noise model remains consistent as the scale increases. However, this assumption often breaks down when the number of qubits changes. Therefore, we explore an alternative quantum error mitigation (QEM) method in the following section.

3.5. Self-mitigation

The self-mitigating (SM) QEM method, suggested in [12], also referred to as decoherence renormalization [48–51], represents a specific error mitigation strategy particularly tailored for Trotterization. The self-mitigation is a special case of the depolarizing noise mitigation method suggested in [52]. The depolarizing noise mitigation method requires a test circuit to measure the noise probability (or noise factor) p . The relation between the true expectation value $\langle O \rangle$ of an observable O and a noisy expectation value $\langle \bar{O} \rangle$ of the observable



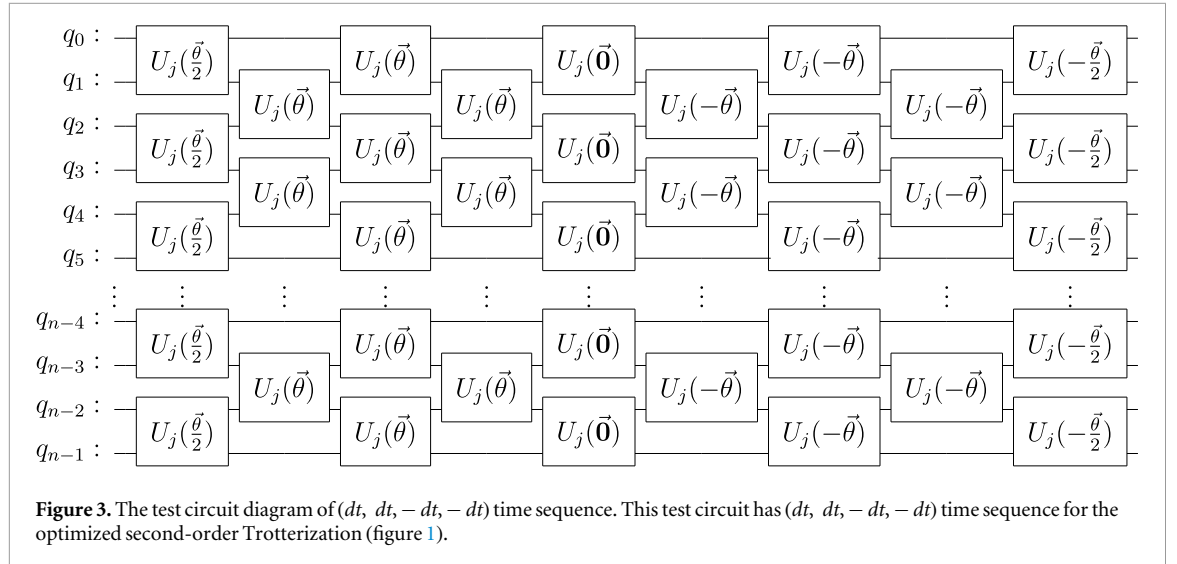
O is described as follows [52]:

$$\langle O \rangle = \frac{\overline{\langle O \rangle}}{1 - p}.$$

where p is the noise factor. For the test quantum circuit measuring $\overline{\langle O \rangle}$, self-mitigation can use the time discretization alternatively in the Trotter step with dt and $-dt$. In other words, the test circuit can have $(dt, -dt, dt, -dt, \dots, dt, -dt)$ time sequence while the target circuit (the original circuit of concern) has $(dt, dt, dt, dt, \dots, dt, dt)$ at the Trotter steps. Since the test circuit applies dt and $-dt$ alternatively in the Trotter steps, the ideal (noiseless) quantum state after the Trotter steps is the initial state, and it is already known or relatively easier to estimate the expectation value of the observable at the initial state. Hence, we have $\langle O \rangle$ by the (ideal) initial state estimation and $\overline{\langle O \rangle}$ by running the test circuit. Then we estimate the noise factor p . Lastly, we run the target Trotterization circuit (forwarding with dt) and obtain the noisy expectation value $\overline{\langle O \rangle}$ of the target circuit. After applying the noise factor p , we estimate the noiseless expectation value $\langle O \rangle$. Note that the discrepancy between the noise factors of the test circuit and the target circuit is negligible to justify the depolarizing noise mitigation method. Since the self-mitigation generates a test circuit with $-dt$ instead of dt , the test circuit preserves the circuit structure of the target circuit. The test circuit has the reversed angle parameter by $-dt$ instead of dt . Hence, we assume the discrepancy between the noise factors of the test circuit and the target circuit is negligible.

We implemented the test circuit of the SM on the optimized second-order Trotterization (refer to figure 1). When we apply $(dt, -dt, dt, -dt, \dots, dt, -dt)$ for the test circuit, the circuit parameters $\vec{\theta}$ in the $U_j(\vec{\theta})$ of even layers become $\vec{0}$ except for the first and the last layer as shown in figure 2, which has $(dt, dt, -dt, -dt)$ time sequence. Since $U_j(\vec{0})$ is the identity operation, all the intermediate operations ($U_j(\vec{0})$) will vanish. In this case, due to the considerable structural mismatch between the test and target circuits, the error rate, p , measured from the test circuit does not accurately reflect that of the target circuit. Therefore, we adopt $(dt, dt, dt, dt, \dots, -dt, -dt, -dt, -dt)$ time sequence for the test circuit. In other words, we apply the dt for the first half sequence and the $-dt$ for the last half sequence. Figure 3 visualizes the test circuit implementation having $(dt, dt, -dt, -dt)$ time sequence. In the $(dt, dt, -dt, -dt)$ time sequence, only the center layer has $\vec{0}$ parameter and the former (the left) has the positive parameters ($\vec{\theta}$ and $\frac{\vec{\theta}}{2}$) and the latter (the right) has the negative parameters ($-\vec{\theta}$ and $-\frac{\vec{\theta}}{2}$). Since only the center layer vanishes, this time sequence keeps a similar structure to the target circuit.

Finally, SM offers several advantages over ZNE. Unlike ZNE, it bypasses the need for extrapolation, thereby avoiding the heuristic assumptions that are a key limitation of ZNE. Even though SM is still heuristic, since the noise factors rely not only on the rotation angle (dt), but also on the accumulated rotations [12], it assumes that the noise factors are not significantly different between the test circuit and the target circuit. Moreover, SM requires only a single test circuit to estimate the noisy observable. In contrast, ZNE typically involves at least two additional circuits to amplify noise, depending on the gate folding scheme. For instance, with a folding schedule of (1, 3, 5), two additional circuits with 3 times and 5 times noise amplification are needed. As a result, SM incurs a constant sampling overhead, making it more resource-efficient.



4. Observable measurement

We analyze the impact of the QEM techniques introduced in section 3 on the time evolution of the XXZ spin chain model and the measurement of the staggered magnetization observable. For the QEM comparison, two types of boundary conditions are employed: the open boundary condition (OBC) and the periodic boundary condition (PBC), both simulated with 20 qubits. After determining the most effective QEM configuration, we extend its application to larger systems with 84 qubits (PBC) and 104 qubits (OBC). For these large-scale cases, we also conduct comparative analyses between the ZNE and SM techniques.

The *direct* approach evaluates the time-evolved expectation value of the staggered magnetization operator $\hat{O}_{M_{st}}$ for a system of N qubits prepared in the Néel state: $\langle \psi(t) | \hat{O}_{M_{st}} | \psi(t) \rangle$, where the state vector evolves as $|\psi(t)\rangle = e^{-iHt} |\psi\rangle$. Here, H represents the Hamiltonian of the XXZ spin chain, expressed as a Hermitian matrix of size $2^N \times 2^N$ acting on a Hilbert space of dimension 2^N . This computation is carried out using the QuSpin library [53]. While the *direct* method provides numerically exact results, it becomes impractical for $N \geq 30$ qubits due to the exponential growth of the Hilbert space. For instance, simulating a 60-qubit system in double precision would require approximately 16 exabytes of memory. To overcome this limitation, we adopt a classical approximation based on matrix product states (MPS) to simulate time evolution when $N > 20$.

MPS offer an efficient representation for studying the time evolution of large quantum many-body systems [54, 55]. An MPS represents the full wavefunction as a one-dimensional array of tensors, each associated with a particular lattice site or spin, and connected through bond indices with maximum dimension χ , known as the bond dimension. Although any quantum state can be represented in MPS form, an exact description typically demands a bond dimension that scales exponentially with system size. In this work, we use the MPS-TDVP method to compute the time-dependent expectation value of the staggered magnetization for the Néel state. This method relies on the time-dependent variational principle (TDVP) [56, 57] and is implemented using the ITensor library [58–60]. The TDVP-based approach supports Hamiltonians with both short- and long-range interactions and exhibits improved computational efficiency for systems with PBC. For the simulations, the time-step is set to $\delta t = 0.1$, with a maximum truncation error of $\epsilon = 10^{-12}$ and a maximum bond dimension $\chi_{\max} = 1000$ per sweep.

In all calculations, we define the time parameter t in units such that $\hbar = 1$ and $J_1 = 1$. To express t in seconds, one can apply the conversion $t \rightarrow \hbar t / J_1$. For a typical magnetic exchange coupling of $J_1 \sim O(\text{eV})$, the corresponding timescale is on the order of 10^{-15} seconds, consistent with atomic transition dynamics.

This section focuses on the dynamical behavior of the XXZ spin chain and, in particular, on evaluating the accuracy of observable measurements on superconducting quantum hardware. We monitor the time-dependent expectation value of the staggered magnetization, an observable that captures the antiferromagnetic ordering of the spin chain, defined as:

$$\hat{O}_{M_{st}} = \frac{1}{N} \sum_{i=1}^N (-1)^i S_i^z. \quad (6)$$

To evaluate $\langle \hat{O}_{M_{st}} \rangle$, we initialize the system in the Néel state, which succinctly characterizes the alternating spin alignment of an antiferromagnet:

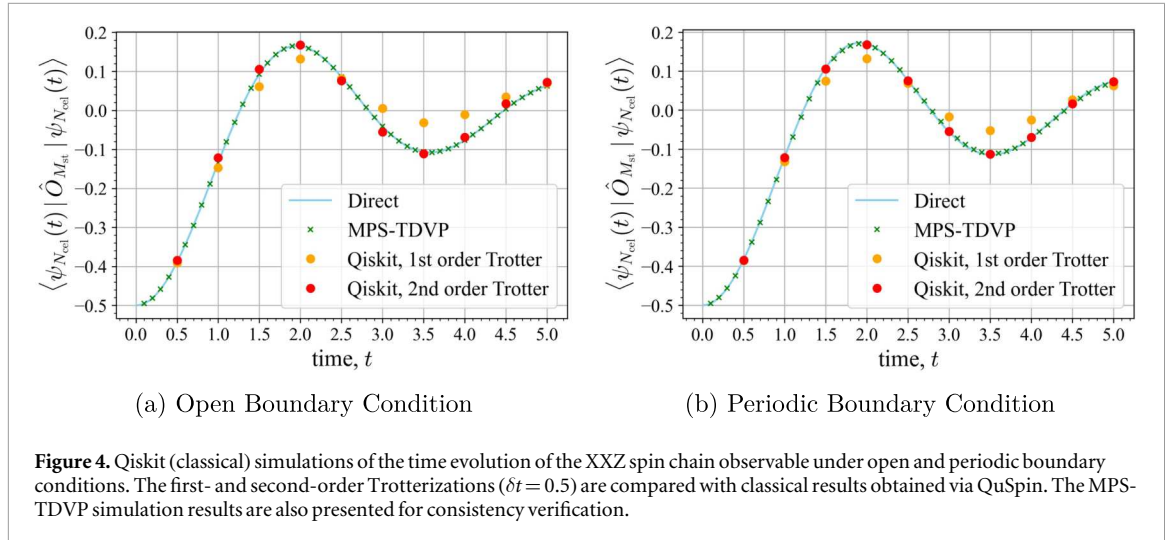


Table 1. Summary of experimental configurations and QEM techniques for system size 20 and boundary conditions.

BC	Qubit Count (N)	QEM Techniques Applied
OBC	20	1. NOQEM; 2. TREX; 3. TREX + DD; 4. TREX + PT; 5. TREX + DD + PT;
PBC	20	1. NOQEM; 2. TREX; 3. TREX + DD; 4. TREX + PT; 5. TREX + DD + PT;

$$|\psi(0)\rangle = |\uparrow\downarrow\uparrow\downarrow\cdots\uparrow\downarrow\uparrow\downarrow\rangle.$$

Here, each $|\uparrow\rangle$ and $|\downarrow\rangle$ denotes the spin-1/2 projection along the z -axis at the i -th lattice site. Using this state as the initial configuration, we compute the time-dependent staggered magnetization both on IBM's superconducting quantum processors and through high-precision classical simulations for validation.

To assess the numerical fidelity of our quantum simulations, we compare the first- and second-order Trotterization methods, as shown in figure 4. A p -th order Trotterization introduces an error of order $O(t(\delta t)^p)$, where t denotes the evolution time and $\delta t = t/r$ corresponds to the time step per Trotter iteration [61]. For a fixed t and r , higher-order decompositions yield smaller Trotterization errors when $\delta t < 1$. Figure 4 illustrates the comparison between first- and second-order Trotterizations using $\delta t = 0.5$ under both OBC and PBC. The second-order scheme exhibits strong agreement with the *direct* simulation, whereas the first-order method deviates noticeably beyond $t = 1.0$. Therefore, the optimized second-order Trotterization [14] with $\delta t = 0.5$ and 100,000 shots is adopted in all subsequent quantum hardware experiments. We further validate the *MPS-TDVP* method against the *direct* computation, observing excellent agreement for both boundary conditions, thereby using *MPS-TDVP* as the benchmark for the $N = 84$ (PBC) and 104 (OBC) simulations.

In the experimental runs, the expectation value of the staggered magnetization for the Néel state under the XXZ Hamiltonian is measured using IBM's *ibm_yonsei* Eagle r3 processor with 127 qubits. Each experiment is executed ten times with 100,000 shots per run. Since the standard deviations across repetitions are small, we plot only the average values in the figures of the experiments in the following sections. The average and standard deviation from ten repetitions are provided for completeness in table A1 and table A4.

4.1. Comparison of TREX, DD, and PT

In order to systematically benchmark the performance of various QEM techniques on 20-qubit systems ($N = 20$) with OBC and PBC, we conducted a series of experiments under configurations summarized in table 1. These include TREX, DD, and PT applied in combinations. The baseline 'NOQEM' case serves as a reference for the effect of the QEM technique.

The comparison results are represented in figures 5, 6, and table A1 for the OBC and the PBC. Figures 5(a) and 6(a) show the time evolution of the observable $\langle \psi_{\text{ex}}(t) | \mathcal{O}_z | \psi_{\text{ex}}(t) \rangle$, with the Qiskit reference (black line) compared against results from NOQEM, TREX, TREX+DD, TREX+PT, and TREX+DD+PT on OBC and PBC, respectively. Figures 5(b) and 6(b) present the absolute error rate compared against the Qiskit reference with OBC and PBC, respectively.

Among the methods, both TREX+DD+PT and TREX+PT closely reproduce the overall shape of the reference curve, aside from amplitude discrepancies in both OBC and PBC cases, with TREX+DD+PT showing

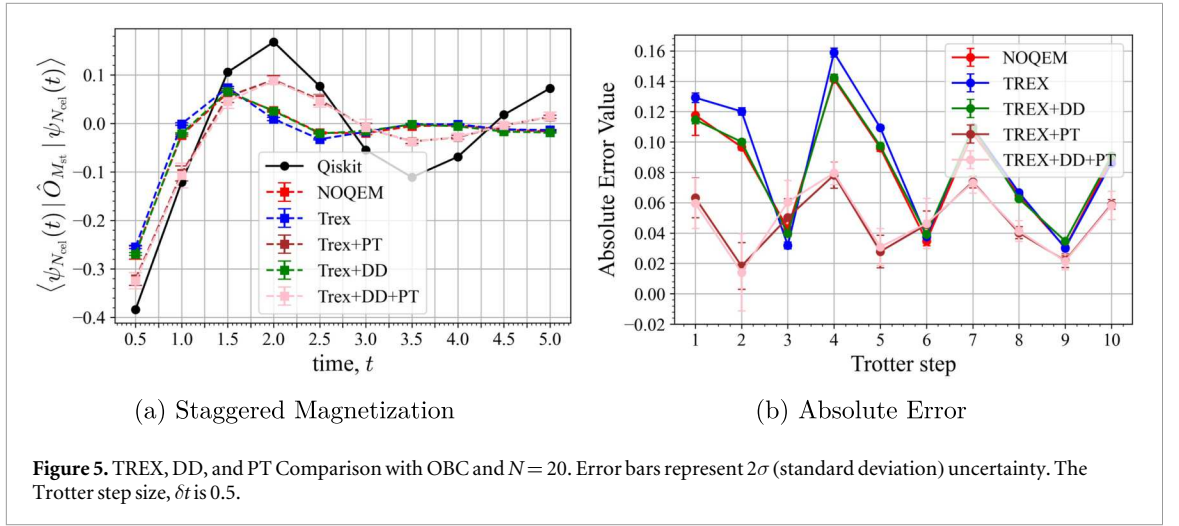


Figure 5. TREX, DD, and PT Comparison with OBC and $N = 20$. Error bars represent 2σ (standard deviation) uncertainty. The Trotter step size, δt is 0.5.

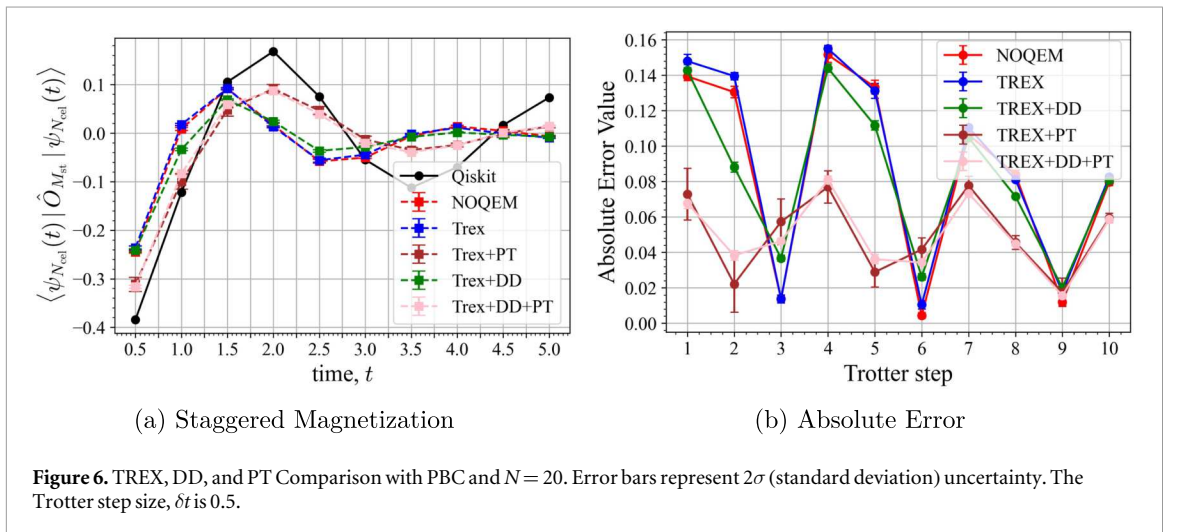


Figure 6. TREX, DD, and PT Comparison with PBC and $N = 20$. Error bars represent 2σ (standard deviation) uncertainty. The Trotter step size, δt is 0.5.

slightly higher accuracy under both boundary conditions. In contrast, results from NOQEM, TREX, and TREX + DD show notable discrepancies with the reference curve at the early time and decay of the observables to zero after time 1.5 (Trotter step 3), likely due to noise-induced distortions by the device error. The quick decay of the observables of NOQEM, TREX, and TREX + DD causes them to align with the reference curve at specific time points incidentally, where the reference values have near-zero values, resulting in temporarily lower absolute errors despite overall poorer performance. This is observed at Trotter steps 3 and 6 in figures 5(b) and 6(b), which present the absolute error at each Trotter step. To mitigate such anomalies and assess overall performance more robustly, the average error values are reported separately in figure 7, clearly indicating the superior accuracy and stability of the TREX + DD + PT strategy across the entire time evolution under both boundary conditions.

The comparison between TREX + DD and TREX + PT in figure 7 shows that the contribution by DD is slight, even though the performance of DD is practically validated in several experiments [62–65]. We conjecture that DD has limited impact in our implementation because the circuit (see figure 1) follows a brick-wall structure, with each brick constructed as described in equation (5). In other words, because our circuit implementation is already dense, there is little room to insert additional DD scheduling. This conjecture is also supported by the PBC case (red histograms) in figure 7. Since the PBC implementation has the brick implementation between the boundary qubits (qubit 0 and 19) in the odd layers, the PBC has less room for the DD scheduling. Hence, the difference between TREX + PT and TREX + DD + PT is negligible. In other words, DD scheduling is unnecessary for the PBC case due to the lack of available space for its insertion.

4.2. Comparison of ZNE and SM

This section extends the QEM comparison to the ZNE and the SM on `ibm_yonsei`. Building on the prior analysis showing that combining TREX, DD, and PT provides the highest accuracy, we now incorporate and

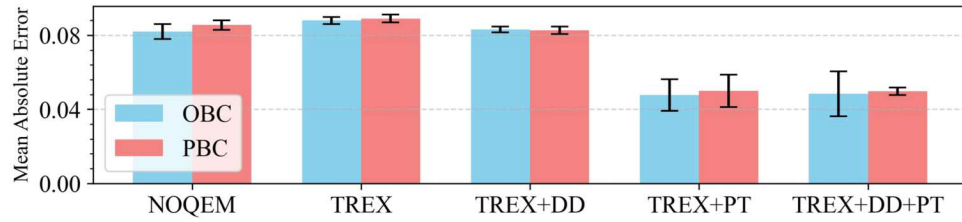
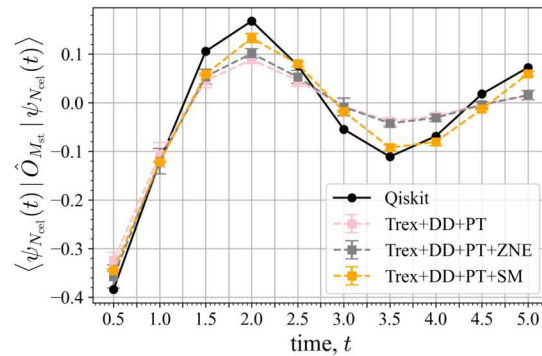
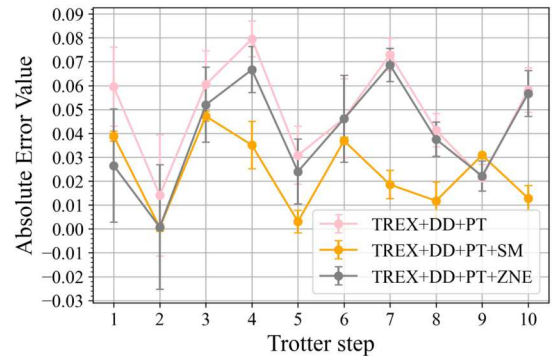


Figure 7. This bar chart presents the mean value of absolute error rates of various Quantum Error Mitigation (QEM) strategies under Open Boundary Conditions (OBC, blue) and Periodic Boundary Conditions (PBC, red) with $N = 20$. Error bars represent 2σ (standard deviation) uncertainty.



(a) Staggered Magnetization for OBC



(b) Absolute Error

Figure 8. Comparison of ZNE and SM with OBC and $N = 20$. Error bars represent 2σ (standard deviation) uncertainty. The Trotter step size, δt is 0.5.

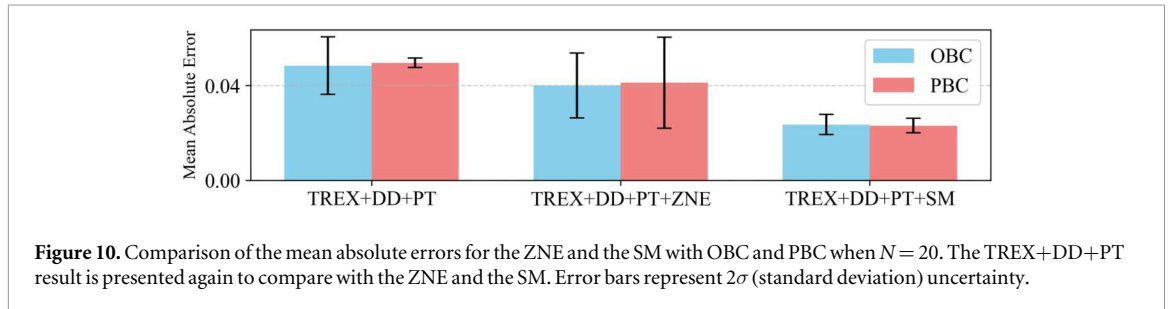
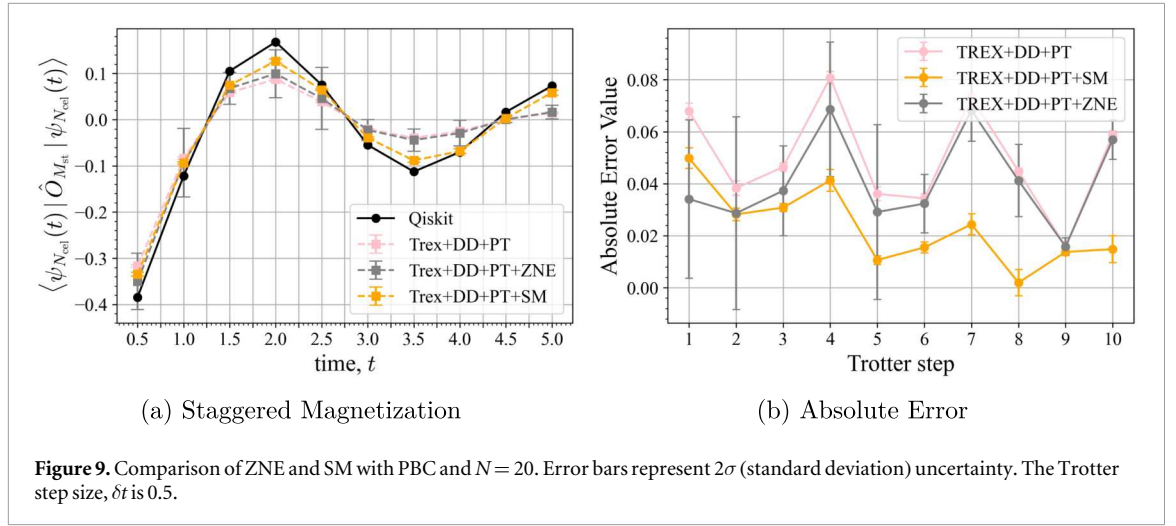
compare ZNE and SM within the TREX+DD+PT framework. The Qiskit results shown in figures 8 and 9 serve as the benchmark for evaluating the absolute error.

As shown in figure 8 (OBC) and figure 9 (PBC), the SM method is more accurate at all the time steps except for the time step 9 of the OBC case (figure 8). At that point, the true value has a near-zero value, and the ZNE results decay to zero after time 3.0 (Trotter step 6), likely due to noise-induced distortions by the device error. Due to the two effects combined incidentally, the ZNE result shows better absolute error than the SM result at the Trotter step 9 of the OBC case. Except for that point, the SM method shows more accurate results than the ZNE method, even though the ZNE always shows a better accuracy than the TREX+DD+PT method, which is the baseline of this comparison.

The corresponding average absolute error rates of these results are summarized in figure 10 and table A3. Specifically, we compare the performance of ZNE and SM, both applied on top of the TREX+DD+PT baseline. The average absolute error value with ZNE is 0.04106 and 0.04123 on OBC and PBC, respectively, whereas the SM has 0.02760 and 0.02218 on OBC and PBC, respectively. The SM achieves an average absolute error reduction of 32.8% for OBC and 46.2% for PBC, respectively. Interestingly, applying SM yields better results than the full combination with ZNE, suggesting that SM's circuit-aware mitigation is more effective than ZNE's heuristic extrapolation under both boundary conditions.

We extend the system size to $N = 104$ for OBC and $N = 84$ for PBC. The PBC system is limited to 84 qubits due to the requirement of connecting the first and last qubits. For the ZNE, we apply the same extrapolation fitting curve used in the $N = 20$ cases. As a benchmark for the true value, we use the MPS-TDVP method, as outlined earlier in this section, since the direct simulation is not feasible for systems of these sizes. The accuracy of the MPS-TDVP is validated in figure 4. These simulations use more than 3, 200 and 2, 600 two-qubit gates on the OBC and the PBC, respectively, as listed in table A2.

Figures 11 and 12 show that the SM method aligns much more closely with the benchmark, represented by green x marks in the figures, than ZNE, even at the larger system sizes of $N = 104$ and $N = 84$. Note that the more accurate results of the ZNE at the Trotter step 9 in both cases are caused incidentally, as explained in the cases of 20-qubit systems (figures 8 and 9). The average absolute errors for each approach are presented in figure 13 and table A3. The average absolute error value with ZNE is 0.04917 and 0.05044 on OBC and PBC, respectively, whereas the SM has 0.02556 and 0.02832 on OBC and PBC, respectively. The SM achieves an



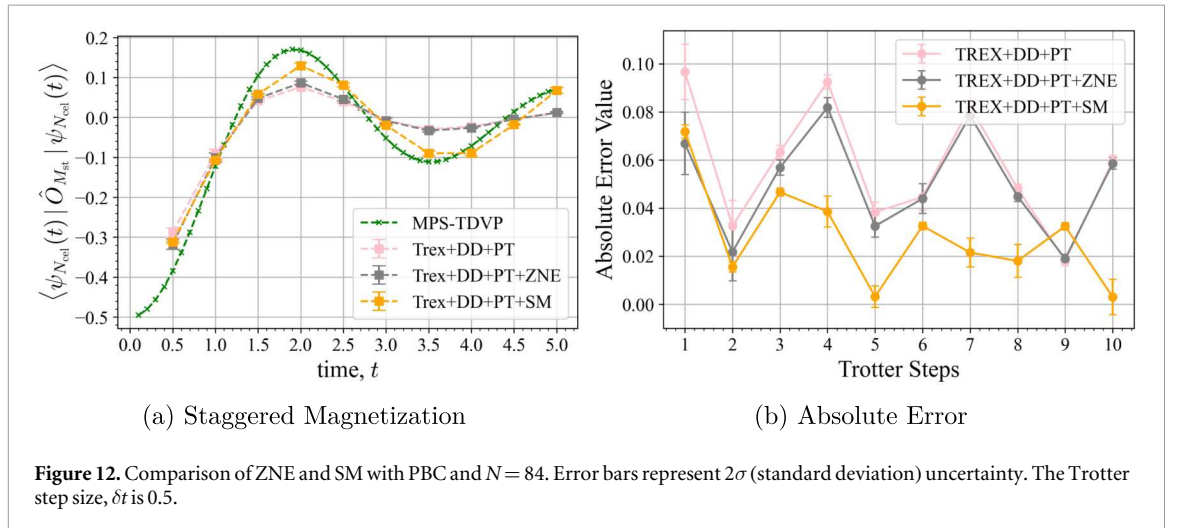
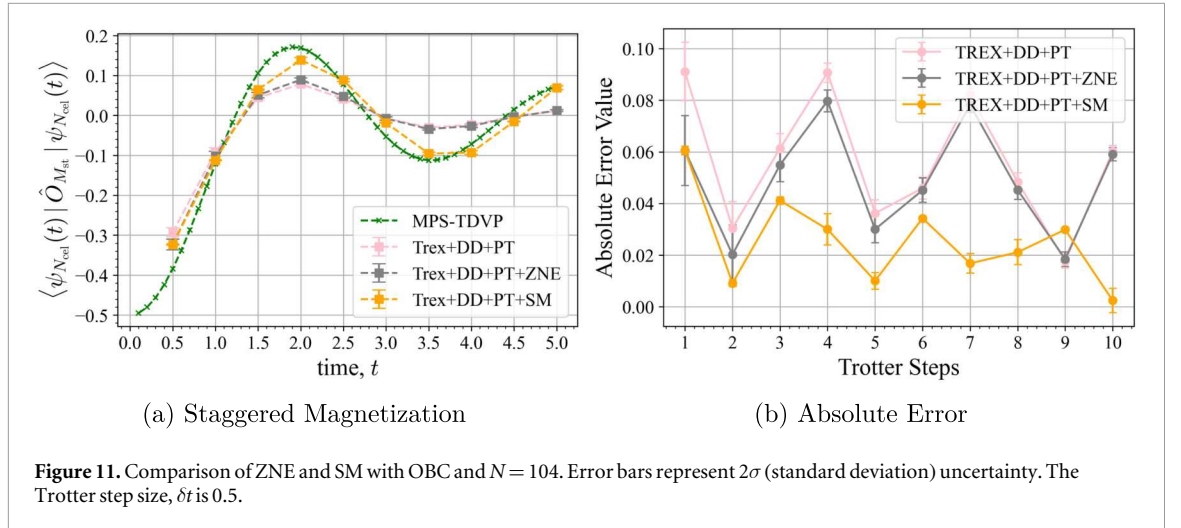
average absolute error reduction of 48.0% on OBC and 43.9% on PBC compared to ZNE. Furthermore, the average absolute errors of ZNE increase by 19.8% and 22.3% compared to the 20-qubit cases for OBC and PBC, respectively.

In conclusion, the SM yields the lowest mean absolute error across all tested system sizes and boundary conditions, clearly outperforming both ZNE and the baseline (TREX+DD+PT) among the three methods. When comparing the 20-qubit systems with the larger systems (104 and 84 qubits), the SM method maintains similar absolute errors, whereas the ZNE method exhibits increased absolute errors in the larger systems. This likely stems from the fact that the ZNE extrapolation curve, derived from the 20-qubit systems, does not accurately capture the error characteristics of the larger systems. In contrast, the SM method more effectively estimates the noise factors in both small and large-scale systems since SM leverages the circuit's inherent noise characteristics and preserves robustness without increasing circuit depth. These features make SM a promising approach for scalable and reliable quantum simulations of Trotterization on near-term noisy quantum devices. However, the SM technique is particularly effective when the true observable can be readily estimated, such as in Trotterization, while ZNE can be applied to more general cases.

5. Entanglement entropy measurement

In this section, we explore measuring entanglement entropy after the time evolution of the Hamiltonian used in section 4 on IBM quantum computers. Measuring entanglement entropy is important because it provides deep insight into the quantum structure and correlations of a system. Entanglement entropy quantifies how strongly different parts of a quantum system are entangled. Unlike classical correlations, entanglement captures nonlocal quantum correlations that are fundamental to quantum mechanics. In particular, in many-body physics, entanglement entropy helps distinguish different quantum phases, such as topological phases, where traditional order parameters fail, and critical phases, where the entropy scales logarithmically with subsystem size.

Despite measuring entanglement entropy being a powerful tool to study quantum systems, it is quite challenging on real quantum computers. Even though we can reconstruct the density matrix of the given system by quantum state tomography (QST), QST is not practical except for very small system sizes because it needs exponential operations with respect to the system size. Hence, we briefly describe a practical entanglement entropy measurement method. When we have a system of N qubits in a pure state $|\psi\rangle$, with its density matrix,



$\rho = |\psi\rangle\langle\psi|$ and we partition this system into two subsystems consisting of L and $M = N - L$ qubits, the entanglement entropy, specifically the Rényi entropy of order n associated with the L -qubit subsystem, is defined as

$$S_L^{(n)} = \frac{1}{1-n} \log [\text{Tr}(\rho_L^n)], \quad (7)$$

where ρ_L is the reduced density matrix for the subsystem with L qubits. This reduced density matrix is obtained by tracing out the subsystem of $M = N - L$ qubits, such that $\rho_L = \text{Tr}_M \rho$. In this study, we focus on the Rényi entropy of order 2 (simply referred to as Rényi entropy from now on). Note that the Rényi entropy of order n converges to the von Neumann entropy:

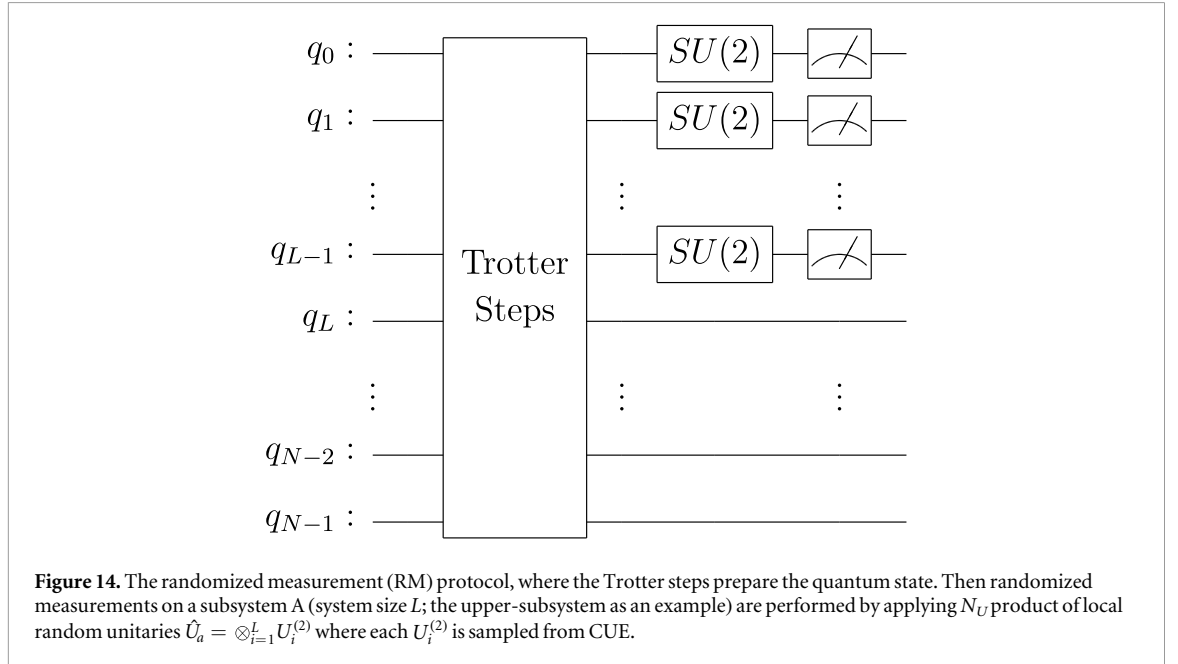
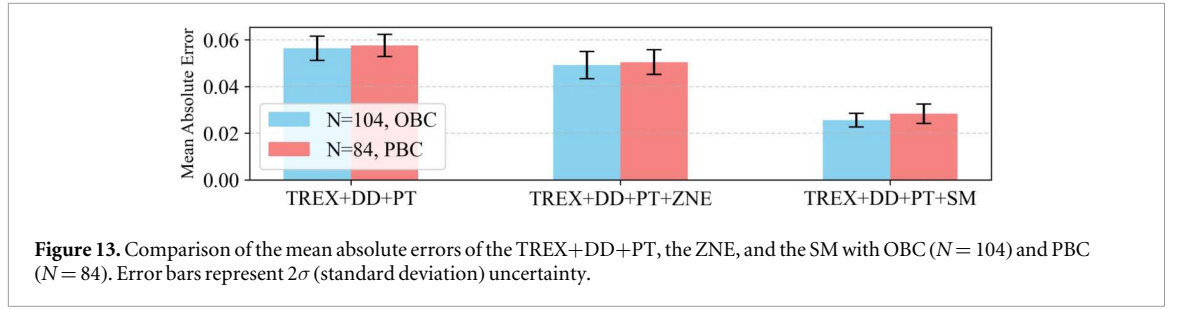
$$S_L^{vN} = -\text{Tr}(\rho_L \log \rho_L)$$

as n approaches 1.

One well-known practical quantum algorithm for the entanglement measurement is the swap-based many-body interference (SWAP-MBI) protocol [66–68] based on the swap test, which is presented in the quantum circuit. When we have an N qubit system, the protocol needs $2N + 1$ qubits because it has two copies of the system and one ancilla (control) qubit. Even though the SWAP-MBI protocol is much more efficient than the QST, it is still limited when the connectivity between qubits is limited, such as in superconducting quantum computers, because the ancilla qubit needs to be connected to all other qubits. Although an efficient implementation of the SWAP-MBI on superconducting quantum computers is proposed [45, 46] on a small system, it is not practical on larger systems. Hence, we adopt the randomized measurement (RM) protocol.

5.1. Randomized measurement protocol

The RM protocol [69–75] leverages the insight that the Rényi entropy of a quantum system can be inferred from statistical correlations among measurement outcomes in randomly chosen bases. In our case, we consider



a pure quantum state ρ of N qubits generated by a random unitary circuit. Consequently, the purity and Rényi entropy are evaluated for the reduced density matrix corresponding to a subsystem A consisting of $L < N$ qubits. A schematic quantum circuit illustrating the RM protocol is shown in figure 14. The experimental realization of this protocol involves several steps, which we outline below.

First, the quantum state ρ is generated by applying the Trotterization steps of N qubits, resulting in a highly entangled state. Our goal is to measure the purity and Rényi entropy of a subsystem A consisting of L qubits. We apply a product of single-qubit unitary operators where each of them acts on the i -th qubit of the A subsystem, i.e., $i \in A$, as

$$\hat{U}_a = U_1^{(2)} \otimes U_2^{(2)} \otimes \dots \otimes U_L^{(2)}.$$

After that, each of the single-qubit unitary $U_i^{(2)}$ is independently sampled from the circular unitary ensemble (CUE) of the $SU(2)$. Afterward, the qubits are measured on a computational basis (i.e., the Z -basis). For each $\hat{U}_a$, repeated measurements are performed to gather statistics (commonly referred to as shots), allowing for the estimation of occupation probabilities, $P_{\hat{U}_a} = \text{Tr}[\hat{U}_a \rho \hat{U}_a^\dagger |j_L\rangle \langle j_L|]$ of computational basis states $|j_L\rangle = |s_1, s_2, \dots, s_L\rangle$ with $s_i = 0, 1$. Note that $\hat{U}_a$ acts only on the subspace of L qubits. Afterward, the entire process is repeated for N_U different randomly drawn instances of $\hat{U}_a$.

Once the set of outcome probabilities $P_{\hat{U}_a}(j_L)$ of the computational basis states $|j_L\rangle$ for one instance of random unitaries, $\hat{U}_a$, the following quantity is computed:

$$X_a = 2^L \sum_{j_L, j'_L} (-2)^{-D[j_L, j'_L]} P_{\hat{U}_a}(j_L) P_{\hat{U}_a}(j'_L)$$

where, j_L and j'_L are different bitstring outcomes for the L qubits, and $P_{\hat{U}_a}(j_L)$, $P_{\hat{U}_a}(j'_L)$ are their corresponding outcome probabilities for the corresponding $\hat{U}_a$. Additionally, $D[j_L, j'_L]$ is the Hamming distance between bitstrings $j_L = s_1 s_2 \dots s_L$ and $j'_L = s'_1 s'_2 \dots s'_L$, measuring how different two bitstrings are, i.e., $D[j_L, j'_L] \equiv \#\{i \in A | s_i \neq s'_i\}$. Consequently the ensemble average of X_a , denoted by $\bar{X}$,

$$\bar{X} = \frac{1}{N_U} \sum_{a=1}^{N_U} X_a \quad (8)$$

is nothing but the second-order cross-correlations across the ensemble of discrete N_U random unitaries $\hat{U}_a$, and provides the estimation of the purity $\text{Tr}(\rho_A^2)$ associated with a subsystem of L qubits, $\{q_{i_1}, \dots, q_{i_L}\}$ where indices $i_1, \dots, i_L$ are chosen from $[0, N - 1]$ in a system of N qubits. Finally, the Rényi entropy (of order 2) is

$$S^{(2)} = -\log \bar{X}.$$

Note that the RM protocol can be recast as the swap-based many-body interference protocol where the swap operator effectively acts on the two virtual copies of ρ_A [73].

5.2. Quantum multi-programming

A central challenge of the RM protocol lies in its dependence on an ensemble of locally applied Haar-random unitaries to suppress variance in entropy estimation. In our experiments, we implement 60 distinct instances of the RM protocol circuits. Additionally, to mitigate errors, we apply TREX, DD, PT, and ZNE. Since applying ZNE and PT repeat each circuit three times (1, 3, 5 gate folding) and ten times for PT (10 PT pair sampling), respectively, in the experimental setting, the total number of circuits executed per the entanglement measurement is 1,800 ($60 \times 3 \times 10$).

Note that only ZNE is applied to this entanglement entropy measurement. The self-mitigation (SM) method has a limitation: it can be used only when the true value (the expectation value in our case) of the test circuit can be easily estimated. This condition is met in the Trotterization test circuit, where the system evolves forward in time by dt and then rewinds by $-dt$. However, the quantum circuit used in the randomized measurement protocol does not meet this condition due to the $SU(2)$ operations at the end of Trotter steps, as shown in figure 14. Therefore, only ZNE is employed for the entanglement entropy measurement.

To make efficient use of the IBMQ system, we employ Quantum Multi-Programming (QMP) to parallelize circuit execution. QMP enables the simultaneous execution of multiple quantum circuits, irrespective of their types or depths. The effectiveness of QMP has been demonstrated across a range of quantum algorithms, including Grover's search, quantum amplitude estimation, the quantum support vector machine, and measuring Rényi entropy [45, 46, 76–78]. Despite its efficiency, QMP introduces certain challenges, such as crosstalk between concurrently executed circuits. To address these issues, we adopt the mitigation strategy described in [76]. In our QMP implementation, we parallelize two circuits per QMP package, inserting one physical idle qubit between them to reduce interference because Ohkura *et al* [79] showed that placing only one physical buffer qubit is sufficient to remove the crosstalk between circuits on IBM quantum computers.

As a result, to execute the full set of 1,800 circuits, we run 900 parallelized circuit pairs. The circuit diagram of the QMP implementation is visualized in figure 15. After executing the QMP circuit, both circuits are measured simultaneously. The measurement outcomes for each circuit are then post-processed following the procedure outlined in appendix of [76].

5.3. Results

The measurement results of the entanglement entropy are shown in figure 16. The Direct results are computed using QuSpin (see section 4 for details on the method). The Qiskit Statevector results are obtained using an optimized second-order Trotterization of the Hamiltonian at the time points where the observable measurements are performed. In this computation, the Rényi entropy is calculated from the statevector after time evolution. The slight discrepancy observed from the fourth time step onward is attributed to Trotterization error. The Qiskit RM Protocol results are generated using the randomized measurement (RM) protocol described in section 5.1, applied after time evolution using the same optimized second-order Trotterization. Hence, this result includes both Trotterization and RM protocol errors. Finally, the `ibm_marrakesh` results are obtained from an IBM Quantum device, `ibm_marrakesh`, which features a Heron r2 processor with 156 qubits. The Qiskit RM Protocol and `ibm_marrakesh` cases use identical circuits to eliminate variability due to circuit differences. As a result, the `ibm_marrakesh` outcome reflects the combined effects of Trotterization error, RM protocol error, and hardware-induced quantum device error. Additionally, the QEM techniques described in section 3 are applied to the `ibm_marrakesh` data.

The Direct, Qiskit Statevector, Qiskit RM Protocol, and `ibm_marrakesh` has 2.07465, 2.11936, 2.04836, and 2.08378, respectively, at time 5.0. The Qiskit Statevector has 2.16% relative error with respect to the Direct. The Qiskit RM Protocol has 3.35% relative error with respect to the Qiskit Statevector. The `ibm_marrakesh` has 1.73% relative error with respect to the Qiskit RM Protocol. Therefore, these results demonstrate that the entanglement entropy measured on the quantum computer aligns well with the noiseless classical simulation.

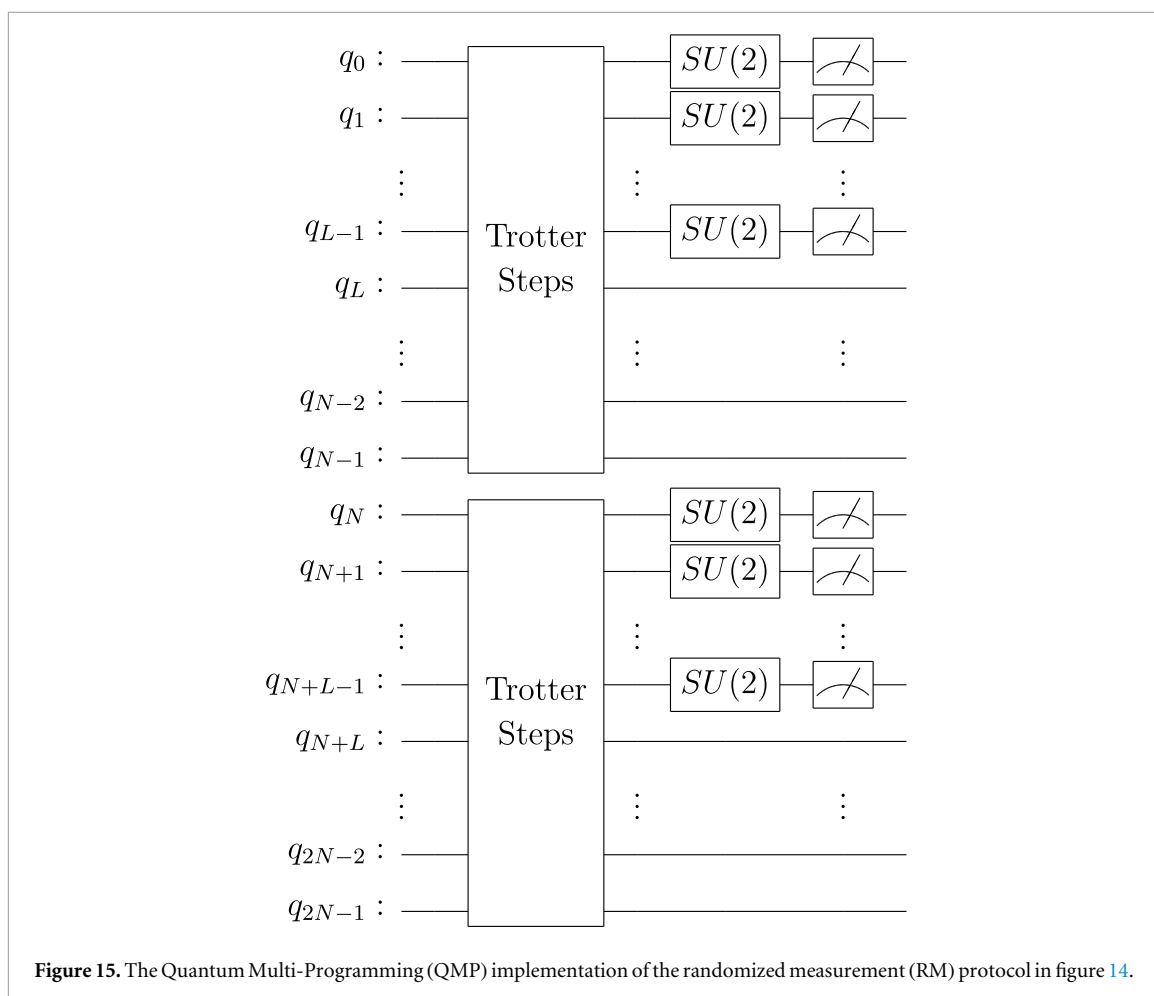


Figure 15. The Quantum Multi-Programming (QMP) implementation of the randomized measurement (RM) protocol in figure 14.

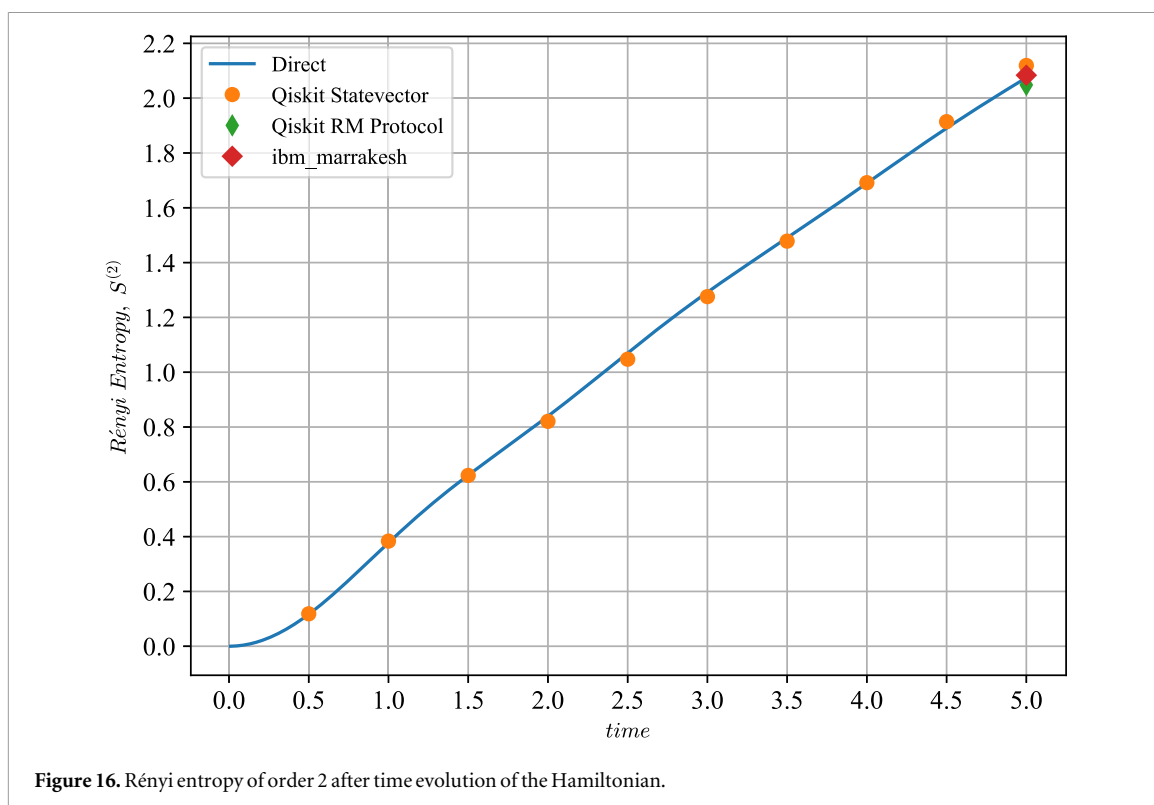


Figure 16. Rényi entropy of order 2 after time evolution of the Hamiltonian.

6. Conclusion and outlook

In this study, we compare a range of QEM methods, examining their characteristics as well as the advantages and limitations of each. Building on this comparative analysis, we assess the capability of current noisy quantum computers to measure observables and entanglement entropy—key quantities for gaining deep insights into quantum systems. We then extend the observable measurements to the quantum utility scale, involving systems with over 100 qubits. The QEM methods examined in this study are relevant to a broad range of quantum computing applications. In particular, TREX, DD, and PT can be applied to general quantum circuits, while ZNE is best suited for problems involving the computation of a small number of observables, such as variational quantum algorithms. The SM method remains less widely used beyond Trotterized simulations, since its application requires that the true value of the test circuit be easily obtainable. In future work, we aim to generalize SM to a wider class of quantum applications.

This research primarily addresses the pre- and early-FTQC stages, but the QEM methods discussed in this paper will continue to offer utility as FTQC systems mature because the mature FTQC is expected to exhibit LER in the range of 10^{-12} to 10^{-15} , which are substantially higher than the machine epsilon error (approximately 2.22×10^{-16} in IEEE 754 double precision) found in classical computers.

The randomized measurement protocol is fundamentally constrained by the high number of measurements needed to reduce statistical errors. Even though an advanced method, such as importance sampling [74], has been developed, the number of randomized measurements with a given accuracy is still of the order of 2^{aN} with $a \approx 1$ when the system has N qubits [71, 75, 80]. We will continue our ongoing efforts to develop more robust and scalable quantum algorithms for measuring entanglement entropy. Despite the limitation of the entanglement entropy, this study will be a significant step forward in the realm of quantum simulation at the utility scale before the fault-tolerant quantum computing era.

The XXZ spin model used in this study adopt only fixed parameters, $J_1 = 1$ and $\Delta = 0.5$. Since the system exhibits a ferromagnetic phase, critical phase, and antiferromagnetic phase when $\Delta < -1$, $-1 < \Delta < 1$, and $\Delta > 1$, respectively, testing different Δ in the different phase will make this work more robust. However, we expect that the parameter variation would not significantly affect the results presented in this study, since it leaves the quantum circuit structure unchanged.

This study expands the applicability of noisy quantum computers to more complex scenarios, specifically addressing large-scale time dynamics of Hamiltonians on noisy superconducting quantum hardware. Looking ahead, our efforts will lay the groundwork for future quantum computing applications, highlighting the quantum advantage over classical methods in simulating complex quantum systems. As we continue to advance the frontiers of quantum computing, our findings add to the growing evidence of its transformative potential in deepening our understanding of quantum phenomena.

Acknowledgments

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Appendix. Experimental data

Table A1. Averaged staggered magnetization of ten repetition with time when $N=20$ on `ibm_yonsei` under NOQEM and TREX+DD+PT conditions and OBC and PBC. The $\pm$ terms represent the standard deviation of the data.

Trotter Steps	OBC		PBC	
	NOQEM	TREX+DD+PT	NOQEM	TREX+DD+PT
1	-0.2387 ± 0.0030	-0.3172 ± 0.0101	-0.2450 ± 0.0012	-0.3165 ± 0.0275
2	-0.0286 ± 0.0009	-0.1107 ± 0.0124	0.0086 ± 0.0016	-0.0836 ± 0.0332
3	0.0710 ± 0.0010	0.0452 ± 0.0067	0.0915 ± 0.0009	0.0588 ± 0.0149
4	0.0121 ± 0.0016	0.0877 ± 0.0029	0.0166 ± 0.0021	0.0873 ± 0.0237
5	-0.0416 ± 0.0025	0.0525 ± 0.0076	-0.0582 ± 0.0020	0.0390 ± 0.0316
6	-0.0363 ± 0.0026	-0.0064 ± 0.0047	-0.0504 ± 0.0008	-0.0205 ± 0.0109
7	-0.0114 ± 0.0018	-0.0349 ± 0.0067	-0.0060 ± 0.0013	-0.0394 ± 0.0115
8	-0.0039 ± 0.0014	-0.0264 ± 0.0043	0.0138 ± 0.0012	-0.0253 ± 0.0131
9	-0.0102 ± 0.0006	-0.0032 ± 0.0025	0.0046 ± 0.0009	0.0008 ± 0.0033
10	-0.0121 ± 0.0004	0.0112 ± 0.0019	-0.0066 ± 0.0010	0.0145 ± 0.0074

Table A2. Circuit depth and the number of CNOT gates.

Trotter Steps	OBC				PBC			
	$N=20$		$N=104$		$N=20$		$N=84$	
	Circuit Depth	CNOTs	Circuit Depth	CNOTs	Circuit Depth	CNOTs	Circuit Depth	CNOTs
1	42	87	42	465	42	90	42	378
2	64	144	64	774	64	150	64	630
3	86	201	86	1083	86	210	86	882
4	108	258	108	1392	108	270	108	1134
5	130	315	130	1701	130	330	130	1386
6	152	372	152	2010	152	390	152	1638
7	174	429	174	2319	174	450	174	1890
8	196	486	196	2628	196	510	196	2142
9	218	543	218	2937	218	570	218	2394
10	240	600	240	3246	240	630	240	2646

Table A3. The mean absolute errors of staggered magnetization under various QEM strategies on `ibm_yonsei`. The results are shown for OBC with $N=20$, $N=104$ and PBC with $N=20$, $N=84$.

QEM Method	OBC		PBC	
	$N=20$	$N=104$	$N=20$	$N=84$
NOQEM	0.08419		0.08550	
TREX	0.08484		0.08896	
TREX+DD	0.08248		0.08267	
TREX+PT	0.05664		0.05002	
TREX+DD+PT	0.04911	0.05641	0.04962	0.05756
TREX+DD +PT+ZNE	0.04106	0.04917	0.04123	0.05044
TREX+DD +PT+SM	0.02760	0.02556	0.02218	0.02832

Table A4. Averaged staggered magnetization of ten repetitions with time when on `ibm_yonsei` under ZNE and SM conditions and OBC ($N = 104$) and PBC ($N = 84$). The $\pm$ terms represent the standard deviation of the data.

Trotter Steps	OBC, $N = 104$		PBC, $N = 84$	
	ZNE	SM	ZNE	SM
1	-0.3237 ± 0.0068	-0.3237 ± 0.0008	-0.3177 ± 0.0065	-0.3127 ± 0.0014
2	-0.1017 ± 0.0057	-0.1128 ± 0.0006	-0.1011 ± 0.0060	-0.1072 ± 0.0010
3	0.0506 ± 0.0033	0.0644 ± 0.0007	0.0476 ± 0.0016	0.0578 ± 0.0008
4	0.0891 ± 0.0021	0.1388 ± 0.0030	0.0862 ± 0.0020	0.1295 ± 0.0032
5	0.0478 ± 0.0026	0.0878 ± 0.0016	0.0454 ± 0.0023	0.0810 ± 0.0022
6	-0.0081 ± 0.0023	-0.0190 ± 0.0004	-0.0085 ± 0.0031	-0.0198 ± 0.0008
7	-0.0344 ± 0.0015	-0.0958 ± 0.0019	-0.0332 ± 0.0020	-0.0901 ± 0.0030
8	-0.0267 ± 0.0019	-0.0931 ± 0.0024	-0.0266 ± 0.0010	-0.0894 ± 0.0034
9	-0.0039 ± 0.0014	-0.0154 ± 0.0004	-0.0047 ± 0.0010	-0.0182 ± 0.0008
10	0.0121 ± 0.0013	0.0688 ± 0.0024	0.0118 ± 0.0012	0.0674 ± 0.0037

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